

RED TEAM vs BLUE TEAM

SECURITY SIMULATION REPORT

REPORT TYPE: FINAL

CANDIDATE: Lakshay

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1. EXECUTIVE SUMMARY

This Final Security Report aggregates all Red Team exploits and Blue Team defense controls performed by Lakshay. It provides an overview of target security posture and mitigations.

2. AUDIT FINDINGS

1. Blue Team Control: parameterized queries

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

2. Blue Team Control: input validation

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

3. Blue Team Control: output encoding

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

4. Blue Team Control: rate limiting

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

5. Blue Team Control: csrf protection

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

6. Blue Team Control: authorization

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

7. Blue Team Control: helmet security

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

8. Blue Team Control: secure cookies

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

9. Blue Team Control: session security

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

10. Blue Team Control: password policies

Severity: INFO | Risk Score: N/A

Mitigation control active in application router.

3. ACTIONABLE RECOMMENDATIONS

1. Control: Maintain Multi-layered Defense-in-Depth

Ensure application filters are configured at the routing layer, controller validation layer, database query layer, and response header layer.

4. SYSTEM CONCLUSION

This audit document lists activities simulated in a secure academic cyber laboratory environment. Operating custom threat models demonstrates that single-point vulnerabilities can compromise entire backend structures. Implementing comprehensive defense-in-depth principles (parameterized inputs, token authorizations, session securities, TLS, security headers) successfully addresses identified attack surfaces and hardens infrastructure boundaries.